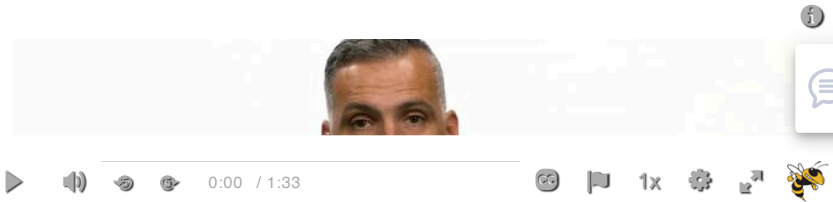


Lesson Ten Overview



Throughout the lecture, we are giving you some **“food for thought”** -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Explain and analyze common models of influence on social networks
- Select optimal “seed nodes” for maximum influence
- Measure cascade phenomena on networks experimentally
- Apply contagion modeling in practice, through case studies

Required and Recommended Reading

Required Reading

- Chapter 19 - D. Easley and J. Kleinberg, [Networks, Crowds, and Markets](https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch19.pdf) , <https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book-ch19.pdf>, Cambridge Univ Press, 2010.

Recommended Reading

- [“The role of social networks in information diffusion”](https://dl.acm.org/doi/abs/10.1145/2187836.2187907) , <https://dl.acm.org/doi/abs/10.1145/2187836.2187907> by E.Bakshy et al. WWW conference, 2012.
- [“Everyone’s an influencer: quantifying influence on Twitter”](https://dl.acm.org/doi/abs/10.1145/1935826.1935845) , <https://dl.acm.org/doi/abs/10.1145/1935826.1935845> by E.Bakshy et al. WSDM conference, 2011.
- [“Maximizing the spread of influence through a social network”](https://dl.acm.org/doi/abs/10.1145/956750.956769) , <https://dl.acm.org/doi/abs/10.1145/956750.956769> by D.Kempe et al. ACM SIGKDD, 2003.
- [“A simple rule for the evolution of cooperation on graphs and social networks”](https://www.nature.com/articles/nature04605) , <https://www.nature.com/articles/nature04605> by H. Ohtsuki et al, Nature, 2006.
- [“Multisensory integration in the mouse cortical connectome using a network diffusion model”](https://www.mitpressjournals.org/doi/abs/10.1162/netn_a_00164?mobileUi=0) , https://www.mitpressjournals.org/doi/abs/10.1162/netn_a_00164?mobileUi=0 by K.Shadi et al., Network Neuroscience, 2020.

Assignments

L10: Knowledge Check

